# Financial Econometrics – Homework 1

Simulation Practice: Simple OLS Regression

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## 1. Setup

Data are generated from $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with $\beta_0 = 5$, $\beta_1 = -2.5$, $x_i \sim U(0,1)$ and $\varepsilon_i \sim N(0, V)$, for three data generating processes: DGP1 ($V = 0.16$, $n = 20$), DGP2 ($V = 1$, $n = 20$) and DGP3 ($V = 1$, $n = 200$). For each sample I build $X = [\mathbf{1} \quad x]$ and compute $\hat{\beta} = (X'X)^{-1}X'y$ with the matrix formula. Each DGP is simulated 10 times with fresh draws (fixed seed for reproducibility). Figure 1 plots data and fitted line for the first run of each DGP; the dashed line is the true line.

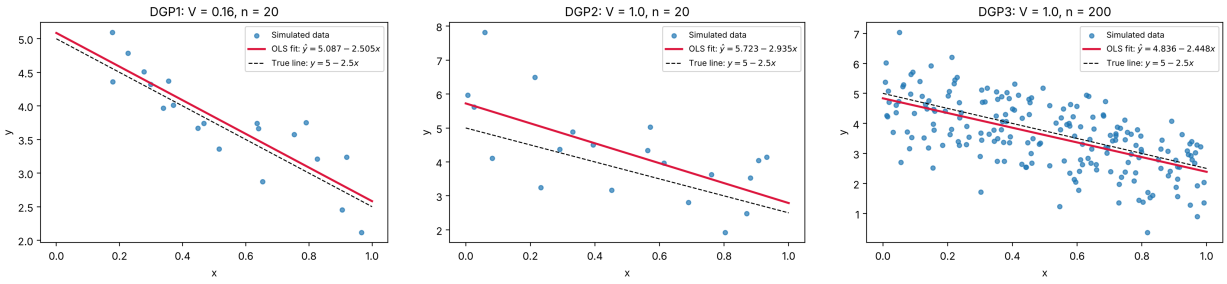



Figure 1: Data and OLS fit, first run of each DGP. Dashed line: true model $y = 5 - 2.5x$.

## 2. Results

Run 1 in Table 1 is the sample shown in Figure 1. The last rows report the average of the 10 estimates, their standard deviation across runs, and the approximate theoretical standard deviation $\sqrt{(V/n) \cdot 4}$ for $\hat{\beta}_0$ and $\sqrt{(V/n) \cdot 12}$ for $\hat{\beta}_1$ (from $\text{Var}(\hat{\beta}) = V(X'X)^{-1}$ with $E[x] = 1/2$, $E[x^2] = 1/3$).

| | DGP1 ($V = 0.16$, $n = 20$) | | DGP2 ($V = 1$, $n = 20$) | | DGP3 ($V = 1$, $n = 200$) | |
|---|---|---|---|---|---|---|
| Run | $\hat{\beta}_0$ | $\hat{\beta}_1$ | $\hat{\beta}_0$ | $\hat{\beta}_1$ | $\hat{\beta}_0$ | $\hat{\beta}_1$ |
| 1 | 5.0865 | −2.5045 | 5.7230 | −2.9348 | 4.8357 | −2.4485 |
| 2 | 4.7899 | −2.2518 | 4.6799 | −2.2719 | 4.6798 | −2.1085 |
| 3 | 4.9072 | −2.4293 | 5.0915 | −2.5309 | 4.7245 | −2.2813 |
| 4 | 4.7564 | −2.1248 | 5.0644 | −3.1262 | 5.1651 | −2.8934 |
| 5 | 4.9386 | −2.3473 | 4.9944 | −3.0375 | 5.0535 | −2.6810 |
| 6 | 4.6632 | −2.1162 | 5.2115 | −2.8415 | 4.8758 | −2.3414 |
| 7 | 5.1338 | −2.8707 | 4.8987 | −2.8717 | 5.0708 | −2.5751 |
| 8 | 4.9058 | −2.0711 | 4.6316 | −1.7000 | 5.4388 | −3.1286 |
| 9 | 4.6653 | −2.1443 | 5.6359 | −3.7015 | 4.7588 | −2.1055 |
| 10 | 4.9624 | −2.2975 | 5.2500 | −2.8237 | 5.0194 | −2.3073 |
| **Mean** | **4.8809** | **−2.3158** | **5.1181** | **−2.7840** | **4.9622** | **−2.4871** |
| Std. dev. (runs) | 0.1616 | 0.2421 | 0.3584 | 0.5333 | 0.2340 | 0.3347 |
| Theoretical std. dev. | 0.1789 | 0.3098 | 0.4472 | 0.7746 | 0.1414 | 0.2449 |
| True value | 5 | −2.5 | 5 | −2.5 | 5 | −2.5 |

Table 1: OLS estimates over 10 repetitions for each DGP.

## 3. Discussion

**(i) Unbiasedness.** In all three DGPs the estimates fluctuate around $\beta_0 = 5$ and $\beta_1 = -2.5$ with no systematic direction of error: OLS is unbiased because $E[\varepsilon \mid x] = 0$.

**(ii) Noise (DGP1 vs. DGP2).** With the same $n = 20$, raising $V$ from 0.16 to 1 makes the estimates much more dispersed: the standard deviation of $\hat{\beta}_1$ across runs rises from 0.24 to 0.53, and single fits can be clearly off (e.g. run 9 of DGP2, $\hat{\beta}_1 = -3.70$). Since $\mathrm{Var}(\hat{\beta}) = V(X'X)^{-1}$, variance grows proportionally to $V$.

**(iii) Sample size (DGP2 vs. DGP3).** With the same $V = 1$, going from $n = 20$ to $n = 200$ reduces the variability: $(X'X)^{-1}$ shrinks roughly like $1/n$, so standard errors fall like $1/\sqrt{n}$ (theoretical std. dev. of $\hat{\beta}_1$: $0.77 \to 0.24$, a factor of about $\sqrt{10}$). In the data, the standard deviation of $\hat{\beta}_1$ falls from 0.53 to 0.33.

**Why.** The estimation error is $\hat{\beta} - \beta = (X'X)^{-1}X'\varepsilon$, a weighted sum of the errors. Larger $V$ makes each $\varepsilon_i$ larger; larger $n$ averages out more errors and increases $X'X$, i.e. the information about the line. So OLS is unbiased in every DGP, and its precision improves with a smaller error variance and a larger sample.

**Caveat on 10 runs.** An average of only 10 estimates is itself random (standard error $\approx$ std. dev.$/\sqrt{10}$), so it should be read as a rough indication. For instance, the DGP1 mean $\hat{\beta}_1 = -2.32$ is 0.18 from the truth, about two standard errors of the mean, which is compatible with chance; likewise, the DGP3 spread across runs (0.23, 0.33) is above the theoretical one (0.14, 0.24), which is normal given that a standard deviation computed from 10 draws is very noisy. As a supplementary check, the code also runs 5000 replications per DGP: the standard deviations of $\hat{\beta}_1$ are 0.326 (DGP1), 0.808 (DGP2) and 0.241 (DGP3), and the means are within about 0.02 of the true values, in line with theory.